

INFO 523

Final

Project

Team 4: Jack Stevens, Rithik
Mehta, Lila Carruthers

Data and Motivation

1. Purpose

Analyze and display AQI and factors influencing it across the United States.

2. Approach

Develop an interactive dashboard with several visualizations and filters.

3. Data

EPA's 2025 air quality database for January through June.

4. Variables

AQI, wind speed arithmetic average, temperature arithmetic average, date, state name, county name

EDA

Cleaning Data

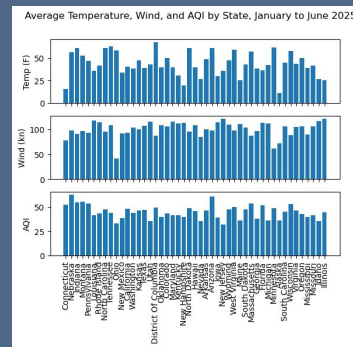
- Dropped columns we would not use
- Renamed columns to be unique to each dataset
 - Example: “method code” became “temp method code” or “wind method code”
- Reordered columns to appear in the same order in all datasets
- Merged 3 datasets together to create master dataset

Final Dataset

State Name	Temp Units of Measure	Wind Units of Measure	
County Name	Temp Observation Count	Wind Observation Count	
Local Site Name	Temp Observation Percent	Wind Observation Percent	AQI
Address	Temp Arithmetic Mean	Wind Arithmetic Mean	AQI Category
Site Num	Temp 1st Max Value	Wind 1st Max Value	AQI Defining Parameter
Longitude	Temp 1st Max Hour	Wind 1st Max Hour	AQI Defining Site
Latitude	Temp Method Code	Wind Method Code	
Date Local	Temp Method Name	Wind Method Name	

Visualization

- Grouped the AQI, mean temperature, and mean wind speed by state name and used the mean value of each
- Converted each to a list
- Created a figure with 3 subplots to compare average values by state



OLS regression model

- The OLS regression model was created to find associations between AQI and the 8 predictors chosen from our temperature and wind datasets.
- Each predictor was evaluated based on relative strength of relationship to AQI
- Value of OLS model is to showcase how each predictor associates to AQI holding all other variables constant

Linear Regression Model

- Model more focused on performance and predictive accuracy
- Uses the same predictor variables with a categorical AQI variable
- Training and testing subsets
- MSE and R squared values given



Results of each Model

- The OLS Regression Model showed multiple significant relationships between our predictor variables and AQI
 - Wind arithmetic mean
 - First max temperature
- Linear regression model showed an R squared value of ~0.60 and MSE of ~137.13
 - Coefficients consistent with OLS regression model
- Overall both models showed significant relationships with specific predictors

OLS Regression Results						
Dep. Variable:	AQI	R-squared:	0.638			
Model:	OLS	Adj. R-squared:	0.638			
Method:	Least Squares	F-statistic:	552.6			
Date:	Tue, 09 Dec 2025	Prob (F-statistic):	0.00			
Time:	16:24:49	Log-Likelihood:	-5.5197e+05			
No. Observations:	111576	AIC:	1.104e+06			
Df Residuals:	111567	BIC:	1.104e+06			
Df Model:	8					
Covariance Type:	nonrobust					
	coef	std err	t	P> t	[0.025	0.975]
const	20.5977	2.354	8.748	0.000	15.983	25.212
Wind Observation Count	0.4785	0.101	4.754	0.000	0.281	0.676
Wind 1st Max Value	0.0102	0.002	6.122	0.000	0.007	0.013
Wind 1st Max Hour	0.0532	0.015	3.520	0.000	0.024	0.083
Wind Arithmetic Mean	-0.0164	0.002	-6.614	0.000	-0.021	-0.012
Temp Observation Count	-0.1944	0.123	-1.581	0.114	-0.435	0.047
Temp 1st Max Value	0.7488	0.025	30.213	0.000	0.700	0.797
Temp 1st Max Hour	0.1760	0.024	7.401	0.000	0.129	0.223
Temp Arithmetic Mean	-0.4885	0.026	-18.461	0.000	-0.540	-0.437
Omnibus:	289642.377	Durbin-Watson:	0.695			
Prob(Omnibus):	0.000	Jarque-Bera (JB):	11677488282.870			
Skew:	29.886	Prob(JB):	0.00			
Kurtosis:	1586.747	Cond. No.	6.12e+03			

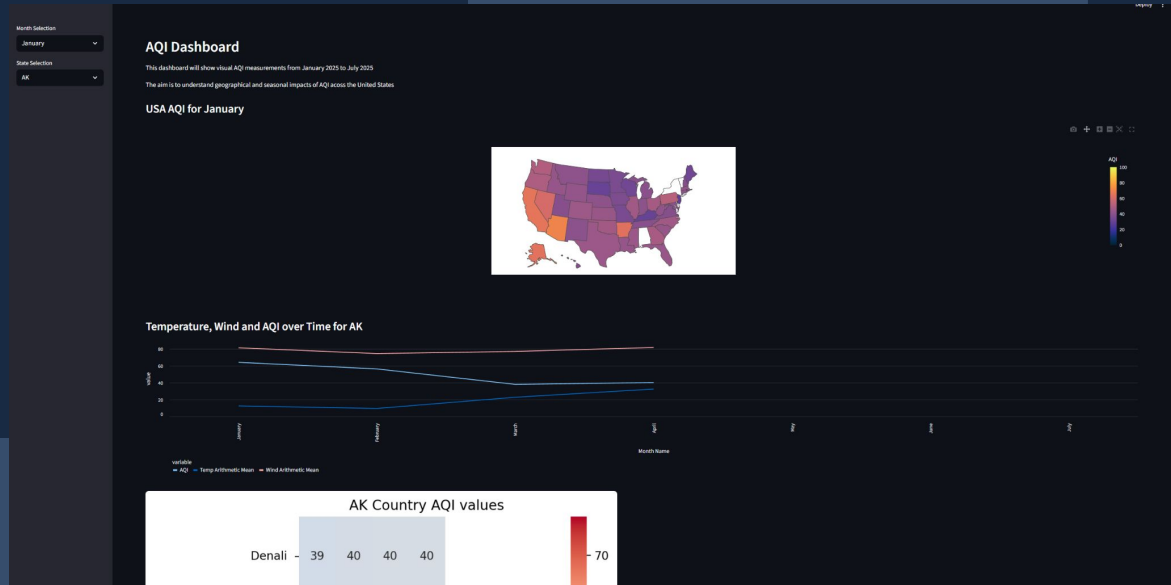
MSE: 137.13117103720404

R²: 0.5993578703493972

Coefficients: [-2.68230539e+01 2.89733524e-01 4.87848810e-03 -1.17293838e-02
-6.64188771e-03 -1.44780982e-01 2.95438874e-01 6.35378437e-02
-2.17573388e-01]

Interactive Dashboard

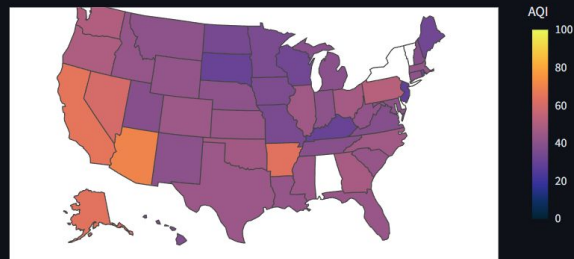
- 3 Key Visuals:
 - Average AQI Heatmap of United States
 - Average AQI Heatmap of State Counties vs Months
 - Trend lines of Average AQI, Average Temperature and Average Wind Speed over Time



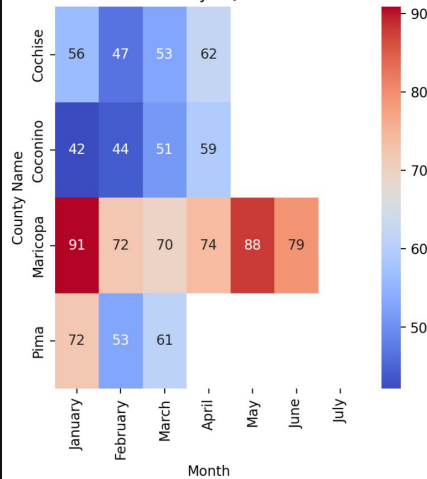
Observations

- Clustering in Southwest States with high AQI values.
- Arkansas significantly higher AQI value than surrounding states.
- Difference between counties in Arizona
 - Could be caused by geological differences

USA AQI for January



AZ Country AQI values



Conclusion

- There are seasonal features that impact the State / County overall Air Quality Index values such as Temperature.
- The dashboard implies that there are geological factors that can be influencing the AQI values.
- There were notable extremes in the US that should be explored with more historical data.
 - Ex. Arkansa

Thank you